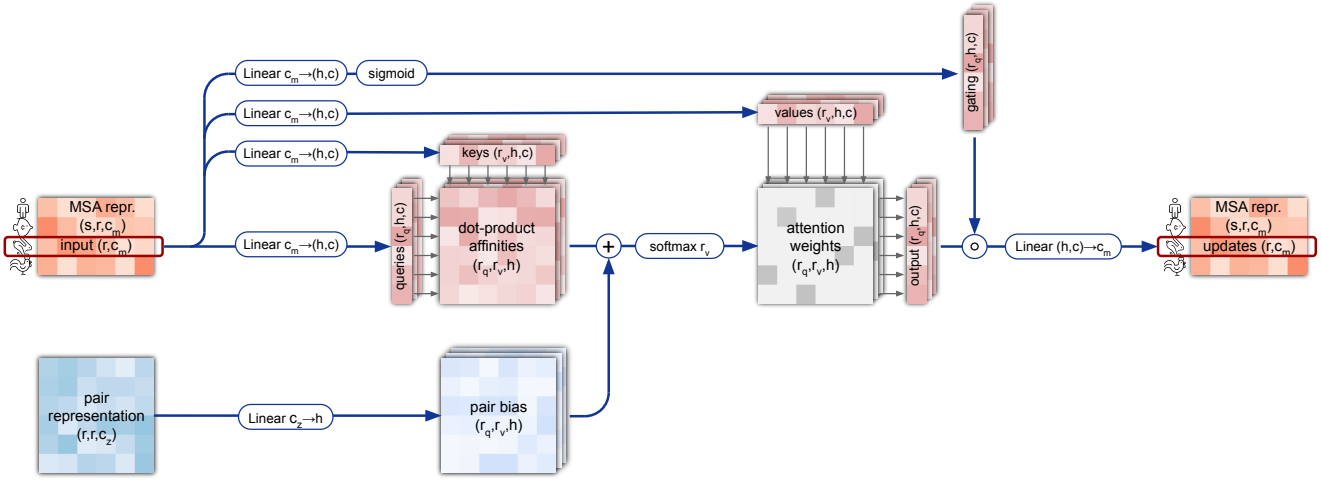


Supplementary Figure 2 | MSA row-wise gated self-attention with pair bias. Dimensions: s: sequences, r: residues, c: channels, h: heads.

Algorithm 7 MSA row-wise gated self-attention with pair bias

def $\text{MSARowAttentionWithPairBias}(\{\mathbf{m}_{si}\}, \{\mathbf{z}_{ij}\}, c = 32, N_{\text{head}} = 8)$:

Input projections

$\mathbf{m}_{si} \leftarrow \text{LayerNorm}(\mathbf{m}_{si})$

$\mathbf{q}^h_{si}, \mathbf{k}^h_{si}, \mathbf{v}^h_{si} = \text{LinearNoBias}(\mathbf{m}_{si})$ $\qquad \mathbf{q}^h_{si}, \mathbf{k}^h_{si}, \mathbf{v}^h_{si} \in \mathbb{R}^c,\ h \in \{1, \ldots, N_{\text{head}}\}$

$b^h_{ij} = \text{LinearNoBias}(\text{LayerNorm}(\mathbf{z}_{ij}))$

$\mathbf{g}^h_{si} = \text{sigmoid}\left(\text{Linear}(\mathbf{m}_{si})\right)$ $\qquad \mathbf{g}^h_{si} \in \mathbb{R}^c$

Attention

$a^h_{sij} = \text{softmax}_j\left(\frac{1}{\sqrt{c}}\, \mathbf{q}^{h\top}_{si} \mathbf{k}^h_{sj} + b^h_{ij}\right)$

$\mathbf{o}^h_{si} = \mathbf{g}^h_{si} \odot \sum_j a^h_{sij} \mathbf{v}^h_{sj}$

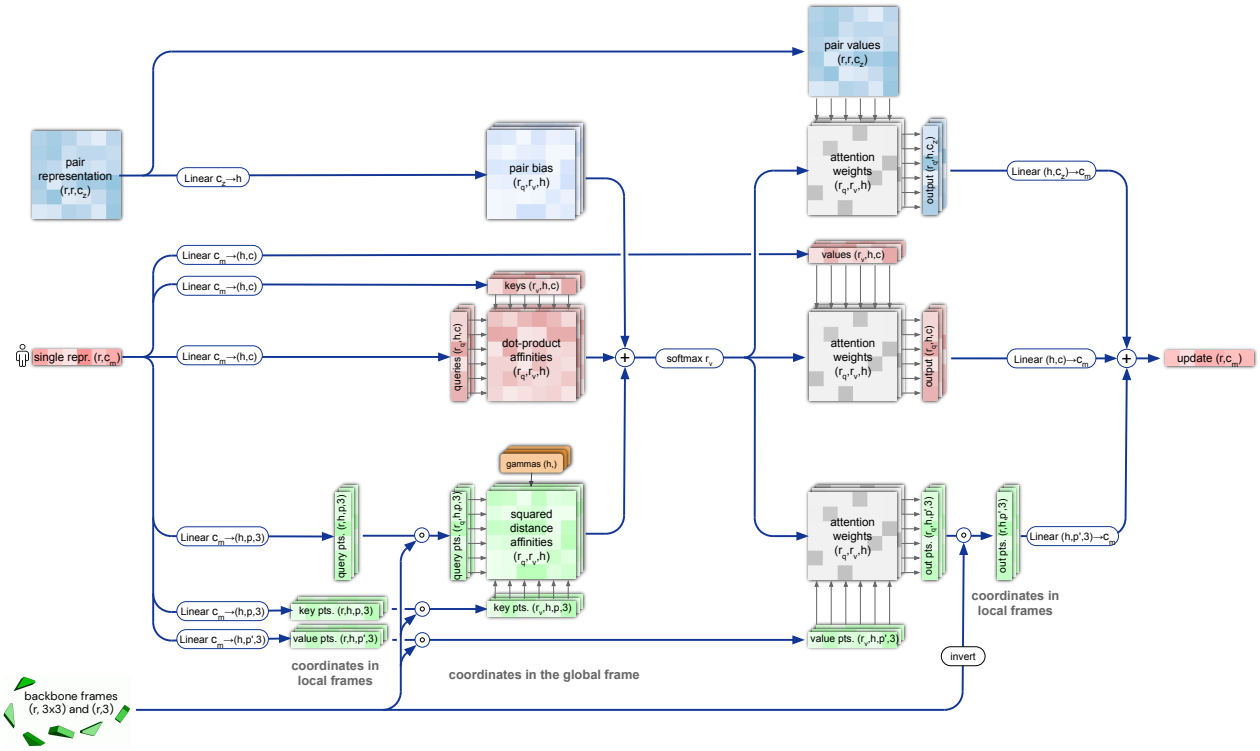
Output projection

$\tilde{\mathbf{m}}_{si} = \text{Linear}\left(\text{concat}_h(\mathbf{o}^h_{si})\right)$ $\qquad \tilde{\mathbf{m}}_{si} \in \mathbb{R}^{c_m}$

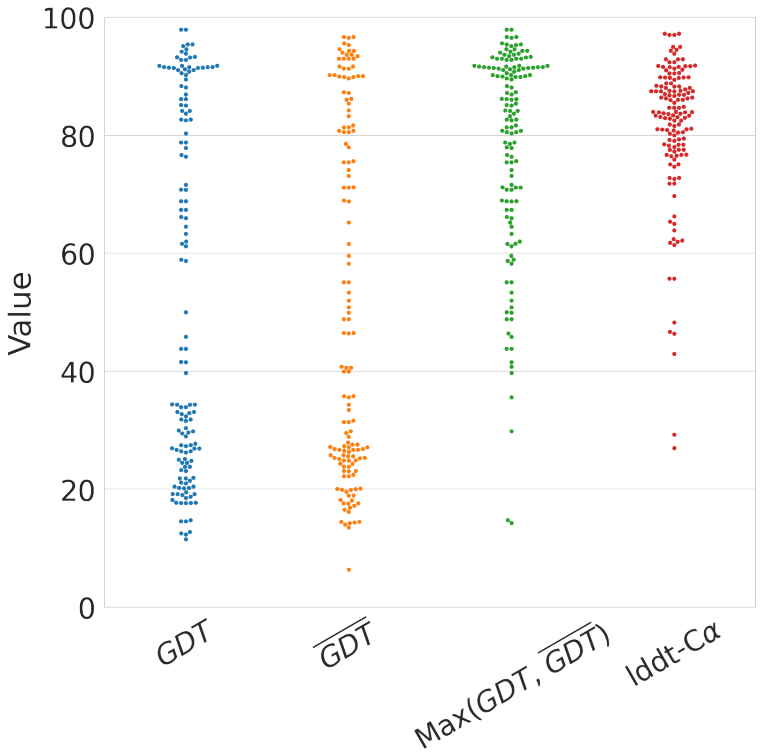
return $\{\tilde{\mathbf{m}}_{si}\}$

1.6.2 MSA column-wise gated self-attention

The column-wise attention (and Suppl. Fig. 3 and Algorithm 8) lets the elements that belong to the same target residue exchange information. In both attention blocks the number of heads $N_{\text{heads}} = 8$, and dimension of the keys, queries, and values $c = 32$.



Supplementary Figure 8 | Invariant Point Attention Module. (**top, blue arrays**) modulation by the pair representation. (**middle, red arrays**) standard attention on abstract features. (**bottom, green arrays**) Invariant point attention. Dimensions: r: residues, c: channels, h: heads, p: points.



Supplementary Figure 9 | Accuracy distribution of a model trained with dRMSD instead of the FAPE loss ($N = 87$ protein domains). Here $\overline{\text{GDT}}$ denotes the GDT score obtained when comparing to the chirally reflected target structure.

since the S function minimizes the quantity in parentheses over all proper rigid transformations and $T_i\tilde{T}_i^{-1}$ is a proper rigid transformation. This inequality simply shows that the point errors averaged over all local frames are no less than the point error for the best single global alignment under the same distance function.

The value $S(\{\vec{\mathbf{x}}_i\}, \{\vec{\tilde{\mathbf{x}}}_i\})$ is zero iff $\text{RMSD}(\{\vec{\mathbf{x}}_i\}, \{\vec{\tilde{\mathbf{x}}}_i\}) = 0$, which shows that $\text{FAPE}(X, \tilde{X}) = 0$ is only possible if the points have $\text{RMSD}(\{\vec{\mathbf{x}}_i\}, \{\vec{\mathbf{y}}_i\}) = 0$. We can characterize all pairs such that $\text{FAPE}(X, \tilde{X}) = 0$ as those where $\text{RMSD}(\{\vec{\mathbf{x}}_i\}, \{\vec{\tilde{\mathbf{x}}}_i\}) = 0$ and $T_i^{-1} \circ \tilde{T}_i$ is one of rigid motions that achieves this zero RMSD (which is unique unless the point sets are degenerate). In particular, the FAPE loss always has non-zero values chiral pairs if the point sets are non-degenerate, regardless of the how the frames are constructed.

1.9.5 Metric properties of FAPE

In this section, we show that idealized FAPE (i.e. using $\epsilon = 0$) has all the properties of a pseudometric, which is a generalized distance function used in topology. To reduce clutter, the proofs are shown without a cutoff but hold equally well when $\|\cdot\|$ is treated as a clipped vector norm. The reader uninterested in the mathematical details may skip this section.

It is simple to show from the defining equation that

$$\text{FAPE}(X, Y) \geq 0 \tag{23}$$
$$\text{FAPE}(X, X) = 0 \tag{24}$$
$$\text{FAPE}(X, Y) = \text{FAPE}(Y, X), \tag{25}$$

so the only remaining pseudometric property to prove is the triangle inequality, $\text{FAPE}(X, Z) \leq \text{FAPE}(X, Y) + \text{FAPE}(Y, Z)$. The triangle inequality is proven below, using tildes and overbars to designate the different con-